Cognitive Infrastructure Failure (CIF)

Definition (working): Cognitive Infrastructure Failure (CIF) occurs when people increasingly rely on external models for interpretation, planning, and meaning—reducing internal model-building and degrading reality-binding over time.

Scope: This framework describes structural failure modes in AI-mediated sense-making environments. It is intended for design, research, and governance discussion.

Exclusions: This document does not assess individual cognitive ability or diagnose psychological conditions.

Core mechanism: Cognition operates through compression: reducing complexity into usable internal representations. When interpretive compression is increasingly externalized to AI systems without sufficient grounding, feedback, or constraint, representational fidelity degrades while outputs remain fluent and coherent.

Failure Modes

1. **Epistemic outsourcing**: External models become the default first-pass interpreter of what is happening, what matters, and what to do.
2. **Internal model degradation**: Reduced ability to form, maintain, and revise internal representations without tool mediation.
3. **Reasoning delegation**: Inferential steps such as judgment, tradeoffs, and stance formation are offloaded rather than supported.
4. **Tool dependence loops**: The tool becomes necessary to recover coherence that prior tool use helped erode, increasing reliance over time.
5. **Interpretive drift**: Internal representations remain coherent but gradually decouple from external reality.

Signatures (Observable)

- Reduced ability to explain a view without paraphrasing a model's output
- "Prompt-first" thinking ("what should I ask?" before "what do I think?")
- Sharp performance or clarity drop when the tool is removed
- Decreased tolerance for ambiguity and faster narrative lock-in
- Preference for coherence over contact with underlying reality
- Fluent output masking declining interpretive fidelity

Simple Tests

Usage: These tests are indicative and comparative, not absolute measures.

1. Solo vs Co-Cognition Transfer Delta

Compare task performance and explanation quality with and without the tool. Large deltas indicate epistemic outsourcing and internal model atrophy.

2. Constraint Retention Across Rewrites

Track whether explicit constraints persist, invert, or degrade across successive model-assisted rewrites.

3. Reversibility Awareness

Ask users to articulate what would change their mind, force revision, or stop a decision. Weak answers indicate feedback decoupling.

4. Source-Contact Ratio

Measure time spent with primary sources versus model-mediated summaries during sense-making.

Design Implications

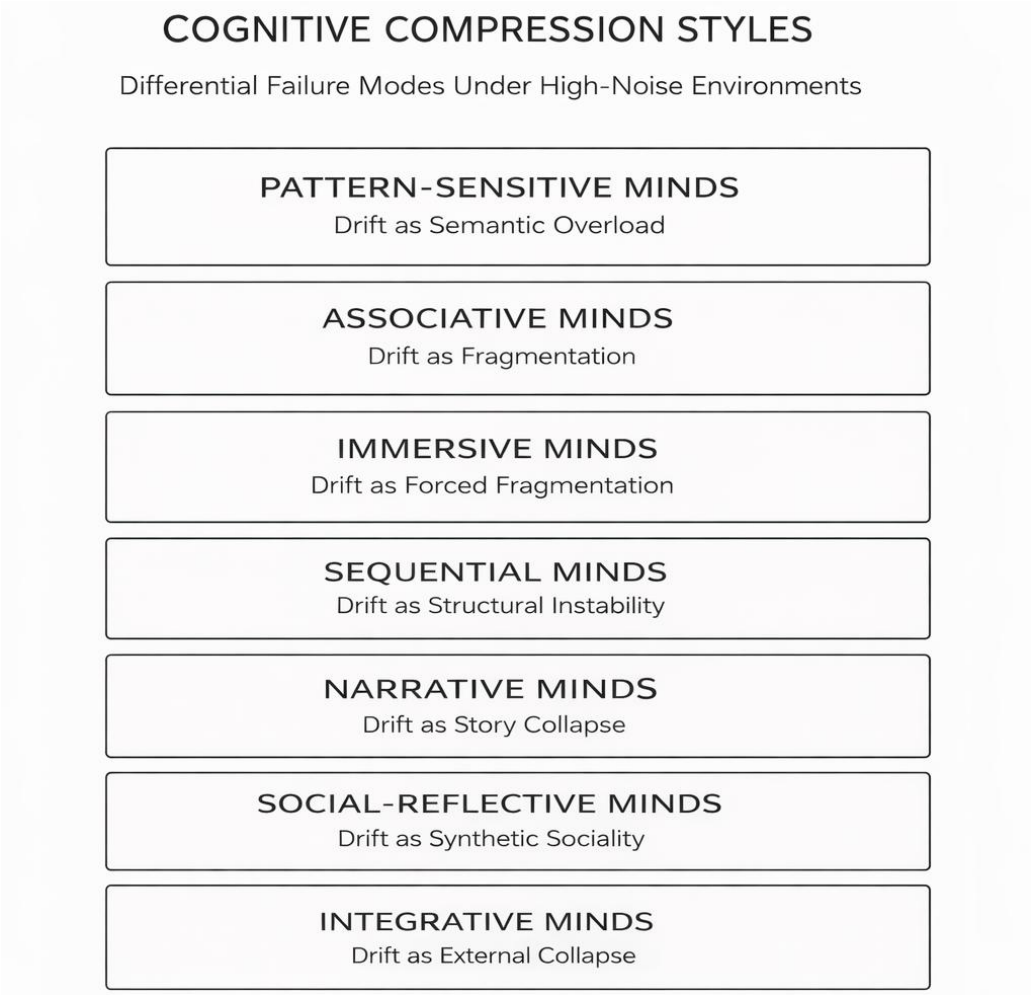
CIF risk increases when systems preserve fluency and continuity while weakening constraint, consequence, or grounding. In these conditions, feedback shifts from corrective to continuative, allowing misalignment to accumulate unnoticed.

Mitigation Patterns

- **Constraint pinning:** Make constraints persistent and binding across interactions
- **Learn-back loops:** Require user-generated explanation before model continuation
- **Uncertainty bands:** Surface ambiguity rather than smoothing it away
- **Grounding re-insertion:** Re-anchor interpretation to primary sources at key stages
- **Tool-off intervals:** Design intentional periods of unaided reasoning

CIF often presents as organizational or system-level stability while imposing increasing orientation and closure costs on individual users. Human cognition evolved in environments where closure was supplied by consequence, resistance, and irreversibility. Not by endless internal refinement.

Cognitive Compression Styles

How different minds drift under constraint collapse and feedback inversion

Cognitive drift emerges when recursive compression operates without stable constraints. As symbolic systems scale, constraint collapse and feedback inversion weaken the stop conditions that once bound thought to consequence. The resulting compression architecture favors fluency and continuation over resolution, producing filter fatigue, explanation without resolution, and runaway recursion. Interpretations continue to refine even after external constraint diminishes, yielding coherent output without closure.